

Wangjia Zhan

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EDUCATION

University of Illinois at Urbana-Champaign <i>Master of Science in Computer Science (MSCS)</i>	Aug. 2025- Present <i>Urbana, USA</i>
University of Illinois at Urbana-Champaign <i>Exchange Student, Major in Computer Science (CS) GPA: 4.0/4.0</i>	Jan. 2024- May 2024 <i>Urbana, USA</i>
The University of Hong Kong <i>Bachelor, Major in Computer Science (CS) cGPA:3.792/4.3 (WES:3.83)</i>	Sep. 2021-Jun. 2025 <i>Hong Kong, China</i>

RESEARCH INTERESTS

Machine learning, Deep learning, Deep learning-based recommendation system (DLRS), Large Language Model, Algorithm Design

PUBLICATIONS

Wangjia Zhan, Haohao Qiu, Bo Min, Lin Lin. Minimal-Cost Route Planning via Fibonacci-Heap-Typed Data Structure. IET Control Theory & Applications. Sep. 2024 [Paper]

RESEARCH EXPERIENCE

Research Assistant <i>Supervised by Prof. Minjia Zhang</i>	Jun 2024 - present <i>Department of Computer Science, UIUC</i>
Project: <i>RecScale: System-Aware Scaling for Large-Scale Deep-learning based recommendation models (DLRMs)</i>	
- Identified fundamental bottlenecks preventing scaling-law behavior in DLRMs, including embedding memory bloat and padding-induced communication inflation in multi-GPU training. - Designed RecScale, an end-to-end system that eliminates memory and communication inflation via embedding compression and distributed communication optimization. - Reduced embedding memory footprint by up to 16× and achieved 3.31× end-to-end training speedup on 64 GPUs, while preserving model accuracy and scaling-law trends.	
Research Assistant <i>Supervised by Prof. James Lam and Doctorate student Lin Lin</i>	Sep. 2023 - Mar. 2024 <i>Faculty of Engineering, HKU</i>
Project: <i>Designing minimal-cost route planning strategies based on advanced data structures and Markov decision process</i>	
- Modeled a route planning problem using Markov Decision Processes (MDPs) and transformed it into a directed graph framework. - Implemented Edmonds' algorithm for initial optimization and further enhanced it with a Fibonacci heap, reducing time complexity from $O(mn)$ to $O(m + n \log n)$. - Designed and presented algorithmic solutions with pseudo-code, effectively bridging theoretical concepts and real-world applications.	

INTERNSHIP EXPERIENCE

Summer Internship - Algorithm Engineer <i>Coached by Professor Zhichao Lu</i>	Jun 2024 - present <i>CityU, HK</i>
Project: <i>Fine-tuning algorithm for Large-scale Models Based on scalable local learning</i>	
- Designed a novel scalable local learning rule leveraging self-supervised learning and data augmentation techniques. - Integrated zero-order optimization methods to enhance the efficiency and adaptability of local learning processes.	
Engineering Summer Internship <i>Coached by Sr. ESE Garin Liu/Supervised by Sr. Reginal Engineering Manager Martin Xu</i>	Jul 2022 - Aug 2022 <i>Microchip Tech, Qingdao</i>
Project: <i>Development of one Embedded Touch Control System with ADC Data Acquisition and Encrypted IrDA Communication</i>	
- Developed and implemented touch key and IR encrypted communication functionalities using Timer, PWM, UART, Configurable Logic Controller (CLC), I2C, and ADC peripherals. - Completed schematic design and PCB layout using Altium Designer. - Debugged and finalized project design using the MPLAB development environment.	

HONORS AND AWARDS

Dean’s Honors List (Faculty of Engineering, HKU)	2024 – 2025
Dean’s Honors List (The Grainger College of Engineering, UIUC)	2024 Spring
Dean’s Honors List (Faculty of Engineering, HKU)	2022 – 2023
Research Assistant Fellowship, Faculty of Engineering, HKU (12,500 HKD)	2023
Overseas Research Fellowship, Faculty of Engineering, HKU (12,000 HKD)	2024

TECHNICAL SKILLS

Programming: Python, C++
Machine Learning: PyTorch, Distributed training, Recommender Systems, Large Language Models, MLsys Efficiency

LANGUAGES

English: Professional, TOEFL:107; GRE:323
Chinese: Native